Linear Algebra and Functional Analysis

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Chapter 1 Linear Algebra (線性代數) and Functional Analysis (泛函分析)

1 Vector Space and Functional

I General

i Vector space (向量空間) or Linear space (線性空間)

A vector space over a field F is a non-empty set V together with a binary operation and a binary function that satisfy the eight axioms listed below. In this context, the elements of V are commonly called vectors, and the elements of F are called scalars.

The binary operation, called vector addition or simply addition assigns to any two vectors v and w in V a third vector in V which is commonly written as $v + w$, and called the sum of these two vectors.

The binary function, called scalar multiplication, assigns to any scalar a in F and any vector v in V another vector in V, which is denoted av.

To have a vector space, the eight following axioms must be satisfied for every u, v and w in V, and a and b in F.

1. Associativity of vector addition: $u + (v + w) = (u + v) + w$.
2. Commutativity of vector addition: $u + v = v + u$.
3. Identity element of vector addition: There exists an element $0 \in V$, called the zero vector, such that $\forall v \in V : v + 0 = v$.
4. Inverse elements of vector addition: $\forall v \in V : \exists$ an element $-v \in V$, called the additive inverse or inverse of v, such that $v + (-v) = 0$.
5. Compatibility of scalar multiplication with field multiplication: $a(bv) = (ab)v$.
6. Identity element of scalar multiplication: $1v = v$, where 1 denotes the multiplicative identity in F.
7. Distributivity of scalar multiplication with respect to vector addition: $a(u + v) = au + av$.
8. Distributivity of scalar multiplication with respect to field addition: $(a + b)v = av + bv$.

When the scalar field is the real numbers, the vector space is called a real vector space, and when the scalar field is the complex numbers, the vector space is called a complex vector space. These two cases are the most common ones, but vector spaces with scalars in an arbitrary field F are also commonly considered. Such a vector space is called an F-vector space or a vector space over F.

When a vector is an n-scalar tuple, we call the scalars components and denote the vector as

$$(x_1, x_2, \dots, x_n)$$

or

$$\langle x_1, x_2, \dots, x_n \rangle$$

ii Linear combination (線性組合)

A linear combination of a series of vectors $(v_1, \dots, v_n)$ in a vector space V over a field F is any vector of the form

$$\sum_{i=1}^{n} c_i v_i$$

where $(c_1, \dots, c_n)$ is a series of scalars in F called coefficients.

iii Isomorphism (同構)

Vector spaces X and Y are called isomorphic if there exists a mapping $f : X \to Y$ between them such that f is bijective and continuous and f^{-1} is continuous, written as $X \cong Y$, and f is called a isomorphism between them.

iv Division

For nonzero vector a and vector b in a vector space such that for some scalar s, $sa = b$, $\frac{b}{a} \coloneqq s$.

v Affine space (仿射空間)

An affine space is a point set A together with a vector space $\vec{A}$, and a transitive and free action of the additive group of $\vec{A}$ on the set A. The vector space $\vec{A}$ is said to be associated to the affine space, and its elements are called vectors, translations, or sometimes free vectors.

Explicitly, the definition above means that the action is a mapping, generally denoted as an addition,

$$A \times \vec{A} \to A$$
$$(a, v) \mapsto a + v$$

that has the following properties.

1. Right identity:
$$\forall a \in A : \ a + 0 = a, \quad \text{where } 0 \text{ is the zero vector in } \vec{A}$$
2. Associativity:
$$\forall v, w \in \vec{A}, \forall a \in A, \ (a + v) + w = a + (v + w)$$

3. Free and transitive action:

$$\forall a \in A : \text{ the mapping } \vec{A} \to A : \ v \mapsto a + v \text{ is a bijection.}$$

4. Existence of one-to-one translations

$$\forall v \in \vec{A} : \text{ the mapping } \vec{A} \to A : \ v \mapsto a + v \text{ is a bijection.}$$

vi Linear map (線性映射) or linear operator

For vector spaces V, W, a map $T : \ V \to W$ between vector spaces V and W over field F is linear if

$$\forall a, b \in F, v, w \in V : \ T(av + bw) = aT(v) + bT(w),$$

also called linear operator.

The space of linear operators from V to W is a vector space, denoted as $L(V, W)$.

vii Linear transformation (線性變換)

A linear transformation is a bijection from an vector space onto itself that is an linear map.

viii Invertible map (可逆映射)

A map $T : \ V \to W$ is Invertible if and only if it is bijective. The unique inverse of T, called T^{-1}, is defined to be:

$$T^{-1} : \ W \to V; \forall v \in V : \ T(v) \mapsto v = T^{-1}(T(v)).$$

ix Linear subspace, vector subspace, or subspace

A linear subspace of a vector space is a subset of it that is closed under vector addition and scalar multiplication, and thus a vector space.

x Linear span, linear hull, or span

A span of a subset S, denoted as span(S) or $\langle S \rangle$, of a vector space is the smallest linear subspace containing S.

xi Linear independence (線性獨立)

A set of nonzero vectors $\{v_1, \ldots, v_n\}$ is linearly independent iff the equation:

$$\sum_{i=1}^{n} c_i v_i = 0$$

is true only if all coefficients $c_1, c_2, \ldots, c_n$ are zero, then the set of vectors is linearly independent; otherwise the set of vectors is linearly dependent.

xii Linearly dependent vector is a linear combination of others

For a linearly dependent set S of nonzero vectors, there exists $v \in S$ such that v is a linear combination of $S \setminus \{v\}$.

Proof. Pick a nonzero c_k. □

xiii Linear independent set as minimum by inclusion of span

A subset X of a vector space is linearly dependent iff there exists a proper subset of X whose span is the span of X.

Proof. If: Let $X = \{x_1, \dots, x_n\}$. There exist not all zero c_i such that

$$\sum_{i=1}^{n} c_i x_i = 0$$

Take any k such that $c_k \neq 0$, we can write x_k in terms of other x_i, thus x_k is in the span of $X \setminus \{x_k\}$, thus $X \setminus \{x_k\}$ is a proper subset of X whose span is the span of X.

Only if: Suppose Y is proper subset of X such that the span of Y is the span of X. Choose $x \in X \setminus Y$. By the definition, there exists $y_1, \dots, y_m \in Y$ such that x is a linear combination of them. □

xiv Basis (基)

The basis of a vector space X over a field F is a set of linearly independent vectors $\mathbf{v} = \{v_1, v_2, \dots\} \subseteq X$ such that for any vector $\mathbf{P} \in X$, there exists a unique coefficient vector $(c_1, c_2, \dots) \subseteq F$ such that:

$$\sum_{i=1} c_i v_i = \mathbf{P}.$$

xv Orientation

The orientation of a vector space is the arbitrary choice of which ordered bases are ”positively” oriented and which are ”negatively” oriented.

xvi Dimension (維度)

The dimension of a vector space V, denoted as $\dim V$, is defined as the cardinality of a basis of it, which is unique (for infinite dimension, if axiom of choice is assumed).

The dimension of an affine space V, denoted as $\dim V$, is defined as the dimension of its associated vector space.

xvii Basis extension theorem

For a vector space V of dimension $n \in \mathbb{N}$, for any linearly independent set of m vectors $v_1, \dots, v_m$ in V with $m \in \mathbb{N}_{\leq n}$, there exists set of $n-m$ vectors $v_{m+1}, \dots, v_n$ in V such that $\{v_1, \dots, v_n\}$ is linearly independent. Moreover, any such $v_1, \dots, v_n$ is a basis of V.

Proof. For $m = n$, $\text{span}\{v_1, \dots, v_n\} = V$, thus $\{v_1, \dots, v_n\}$ is a basis. For $m < n$, since $S \coloneqq \text{span}\{v_1, \dots, v_m\}$ is of dimension $m < n$, choose any $v_{m+1} \in V \setminus S$. Assume for contradiction that $\{v_1, \dots, v_{m+1}\}$ is linearly dependent, we can write v_{m+1} as a linear combination of $v_1, \dots, v_m$, which means $v_{m+1} \in S$. Repeat until n. □

xviii Bilinear form (雙線性形式)

Let V be a vector space of dimension n over a field $\mathbb{K}$. A map $B : V \times V \to \mathbb{K}$ is a bilinear form on the space if:

1. $\forall u, v, w \in V : B(u + v, w) = B(u, w) + B(v, w)$.
2. $\forall u, v \in V, \lambda \in \mathbb{K} : B(\lambda u, v) = \lambda B(u, v)$.
3. $\forall u, v, w \in V : B(u, v + w) = B(u, v) + B(u, w)$.
4. $\forall u, v \in V, \lambda \in \mathbb{K} : B(u, \lambda v) = \lambda B(u, v)$.

xix Symmetric bilinear form (對稱雙線性形式)

Let V be a vector space of dimension n over a field $\mathbb{K}$. A map $B : V \times V \to \mathbb{K}$ is a symmetric bilinear form on the space if:

1. $\forall u, v \in V : B(u, v) = B(v, u)$.
2. $\forall u, v, w \in V : B(u + v, w) = B(u, w) + B(v, w)$.
3. $\forall u, v \in V, \lambda \in \mathbb{K} : B(\lambda u, v) = \lambda B(u, v)$.

xx Topological vector space (TVS) (拓樸向量空間)

A topological vector space X is a vector space over a topological field $\mathbb{K}$, such that vector addition $X \times X \to X$ and scalar multiplication $\mathbb{K} \times X \to X$ are continuous functions.

xxi Functional (泛函)

A functional is a function from a vector space into the field of real or complex numbers. For a vector space X, $x \in X$ and a functional on f on X, $f(x)$ is also denoted as $\langle f, x \rangle$ or fx.

xxii Seminorm (半範數)

Given a vector space X over a ordered field F, a seminorm on X is a real-valued function $p : X \to \mathbb{R}$ with the following conditions, where $|s|$ denotes the usual absolute value of a scalar s:

1. Subadditivity/Triangle inequality:

$$\forall x, y \in X : p(x + y) \leq p(x) + p(y).$$

2. Absolute homogeneity:

$$\forall s \in \mathbb{R}, x \in X : p(sx) = |s| p(x).$$

These conditions implies that:

1. Non-negativity: $\forall x \in X :\ p(x) \geq 0$.
2. $p(0) = 0$.

xxiii Norm (範數/模長)

A norm on X is a seminorm $p :\ X \to \mathbb{R}$ with the following properties:

- Positive definiteness/Positiveness/Point-separating:

$$\forall x \in X :\ p(x) = 0 \implies x = 0$$

A vector x such that $p(x) = 1$ is called an unit vector.

xxiv Sublinear function (亞線性函數) or quasi-seminorm (準半範數)

Let $p :\ X \to \mathbb{R}$ be a function on a vector space X over the field $\mathbb{K}$, which is either $\mathbb{R}$ or $\mathbb{C}$. p is called a sublinear function or quasi-seminorm if it satisfies the following conditions:

1. Subadditivity/Triangle inequality:

$$\forall x, y \in X :\ p(x + y) \leq p(x) + p(y).$$

2. Nonnegative homogeneity:

$$\forall x \in X, r \geq 0 :\ p(rx) = rp(x).$$

xxv Operator norm or induced norm

For normed vector spaces V, W, the vector space of linear operators $L(V, W)$ is equipped with a norm $\|\cdot\|_L$ called operator norm or induced norm defined as, for $A \in L(V, W)$,

$$\|A\|_L = \sup_{\|v\|_V \leq 1} \|A(v)\|_W.$$

xxvi Absolutely continuous function (絕對連續函數)

Function $f :\ I \subseteq \mathbb{R} \to \mathbb{R}$ is absolutely continuous on I if for every positive number ε, there is a positive number δ such that, for every finite sequence of pairwise disjoint sub-intervals (x_k, y_k) of I with $x_k < y_k \in I$ and cardinality N,

$$\sum_{k=1}^{N} (y_k - x_k) < \delta \implies \sum_{k=1}^{N} |f(y_k) - f(x_k)| < \varepsilon.$$

The collection of all absolutely continuous functions on I is denoted $\mathrm{AC}(I)$.

xxvii Bounded linear operator (有界線性運算子)

A bounded linear operator is a linear transformation $L : X \to Y$ between topological vector spaces X and Y that maps bounded subsets of X to bounded subsets of Y. If X and Y are normed vector spaces, then L is bounded if and only if there exists some $M > 0$ such that:

$$\forall x \in X : \|Lx\|_Y \leq M\|x\|_X.$$

The smallest such M is called the operator norm of L and denoted by $\|L\|$. A bounded operator between normed spaces is continuous and vice versa.

xxviii Uniform norm, sup norm, supremum norm, Chebyshev norm, infinity norm, or max norm

Uniform norms are defined on the function space of bounded functions valued in a normed space. The uniform norm of bounded function $f : X \to (Y, \|\|_Y)$ is defined by

$$\|f\|_\infty = \sup_{x \in X} \|f(x)\|_Y.$$

xxix Hahn–Banach theorem (哈恩-巴拿赫定理)

Let $p : X \to \mathbb{R}$ be a sublinear functional on a vector space X over the field $\mathbb{K}$, which is either $\mathbb{R}$ or $\mathbb{C}$. If $f : M \to \mathbb{K}$ is a linear functional on a vector subspace M such that

$$\forall m \in M : f(m) \leq p(m),$$

then there exists a linear functional $F : X \to \mathbb{K}$ such that

$$\forall m \in M : F(m) = f(m),$$

$$\forall x \in X : F(x) \leq p(x).$$

xxx p-norm (p-範數)

The p-norm of a n-dimensional vector $x = ((x_i)_{i=1}^n)$ is defined to be

$$\|x\|_p = \left(\sum_{i=1}^{n} |x_i|^p \right)^{\frac{1}{p}}, \quad p \in \mathbb{N}_\infty,$$

where the ∞-norm is

$$\|x\|_\infty = \max_{i=1}^{n} |x_i|.$$

The 1-norm is also called Manhattan norm (曼哈頓範數); the 2-norm is also called Euclidean norm (歐幾里德範數).

xxxi Locally convex topological vector space (LCTVS) or locally convex space

A locally convex topological vector space (LCTVS) $(X, \mathcal{T})$ is a TVS whose topology is generated by a family of seminorms P on it such that:

$$\bigcap_{p \in P} \{x : p(x) = 0\} = \{0\}$$

That is, $\mathcal{T}$ is generated by a basis $\{U_{\epsilon,x}\}_{\epsilon \in \mathbb{R}_{\geq 0}, x \in X}$ of neighborhoods defined to be:

$$U_{\epsilon,0} = \{y \in X : \forall p \in P : p(y) < \epsilon\}.$$

$$U_{\epsilon,x} = \{y \in X : y - x \in U_{\epsilon,0}\}.$$

This definition implies that a LCTVS is necessarily a Hausdorff space (T2 space).

xxxii Banach space (巴拿赫空間)

A Banach space is a normed vector space $(X, \|\cdot\|)$ with induced metric d

$$d(x, y) = \|x - y\|, \quad \forall x, y \in X$$

that is a Complete metric space (aka Cauchy space).

xxxiii Dot product (點積)

The dot product of two vectors $\mathbf{a} = (a_1, a_2, \ldots, a_n)$ and $\mathbf{b} = (b_1, b_2, \ldots, b_n)$ is defined as:

$$\mathbf{a} \cdot \mathbf{b} = \sum_{i=1}^{n} a_i b_i.$$

Two nonzero vector are called orthogonal or perpendicular iff their dot product is zero.

xxxiv Inner product space (內積空間)

A bar over a scalar denotes the complex conjugate of it. An inner product space is a vector space V over the real or complex field F together with an inner product, that is, a map

$$\langle \cdot, \cdot \rangle : V \times V \to F$$

that satisfies the following three properties for all vectors $x, y, z \in V$ and all scalars $a, b \in F$.

- Conjugate symmetry:

  $$\langle x, y \rangle = \overline{\langle y, x \rangle}.$$

  This implies that $\langle x, x \rangle$ is a real number.

- Linearity in the first argument:

  $$\langle ax + by, z \rangle = a\langle x, z \rangle + b\langle y, z \rangle.$$

- Positive-definiteness: if x is not zero, then $\langle x, x \rangle > 0$.

Every inner product space has an induced norm called canonical norm and is often assumed to be equipped defined by

$$\|x\| = \sqrt{\langle x, x \rangle}$$

The inner product space over $\mathbb{C}$ is also called unitary space.

xxxv Inner product is bilinear form

An inner product is a bililinear form. TODO

Proof. TODO □

xxxvi Orthonormal basis (正交基)

An orthonormal basis for an inner product space V with finite dimension is a basis for V such that for e, f in it:

$$\langle e, f \rangle = \begin{cases} 1, & \text{if } e = f, \\ 0, & \text{if } e \neq f. \end{cases}$$

xxxvii Balanced set (平衡集), circled set, or disk set

Let X be a vector space over a field $\mathbb{K}$ with an absolute value function $|\cdot|$. A subset S of X is called a balanced set or balanced if:

$$\forall a \in \mathbb{K} \text{ s.t. } |a| \leq 1 : \ aS \subseteq S.$$

xxxviii Well-defined (定義良好)/Unambiguous (不模糊的)

A well-defined expression or unambiguous expression is an expression whose definition assigns it a unique interpretation or value. Otherwise, the expression is said to be not well defined, ill defined or ambiguous.

xxxix Affine map (仿射映射)

Given two affine spaces A and B whose associated vector spaces are $\mathscr{A}$ and $\mathscr{B}$, an affine map from A to B is a map $f : A \to B$ such that

$$\begin{aligned} f &: \mathscr{A} \to \mathscr{B} \\ b - a &\mapsto f(b) - f(a) \end{aligned}$$

is a well-defined linear map.

xl Affine transformation (仿射變換)

An affine transformation is a bijection from an affine space onto itself that is an affine map.

xli Von Neumann bounded or bounded

Suppose X is a topological vector space over a field $\mathbb{K}$. A subset B of X is called von Neumann bounded (or bounded) in X if for every neighborhood V of the origin, there exists a real $r > 0$ such that $B \subseteq sV$ for all scalars s satisfying $|s| \geq r$.

xlii Convex set (凸集)

Let S be a vector space or an affine space over some ordered field. A subset C of S is convex if, for all x and y in C, the line segment connecting x and y is included in C.

xliii (Algebraic) dual (對偶) (vector) (space)

Given any vector space V over a field F, the dual space V^* is defined as the set of all linear maps $\varphi : V \to F$. The dual space V^* itself thus becomes a vector space over F when equipped with an addition and scalar multiplication satisfying that for any $\varphi, \psi \in V^*$, for any $x \in V$, for any $a \in F$,

$$(\varphi + \psi)(x) = \varphi(x) + \psi(x),$$

$$(a\varphi)(x) = a\,(\varphi(x))\,.$$

Element in dual is called covector.

xliv Continuous dual (space)

Given any vector space V over a field F, the continuous dual space V' is defined as the set of all continuous linear maps $\varphi : V \to F$. The continuous dual space V' itself thus becomes a vector space over F when equipped with an addition and scalar multiplication satisfying that for any $\varphi, \psi \in V'$, for any $x \in V$, for any $a \in F$,

$$(\varphi + \psi)(x) = \varphi(x) + \psi(x),$$

$$(a\varphi)(x) = a\,(\varphi(x))\,.$$

xlv Transpose, dual map, or adjoint

For two vector spaces X, Y and a continuous linear map $A : X \to Y$, the transpose, dual map, or adjoint of A is defined as the unique map ${}^tA : Y' \to X'$ such that for all $x \in X$, for all $y \in Y'$,

$$\langle y, A(x)\rangle = \langle {}^tA(y), x\rangle.$$

xlvi Strong dual (space)

A strong dual space is a continuous dual space of a TVS equipped with the strong (dual) topology, also called topology of uniform convergence on bounded subsets. Suppose X is a TVS over field $\mathbb{F}$ and $\mathcal{B}$ is a family of bounded subsets of X such that every bounded subset of X is a subset of some $B \in \mathcal{B}$. Define the family T of seminorms $\|\cdot\|_B$ on X' for all $x' \in X'$ as

$$T = \{\|x'\|_B \coloneqq \sup_{x \in B} \left|x'(x)\right| \mid B \in \mathcal{B}\}.$$

The strong dual topology is defined as the topology generated by T, and thus B' is a LCTVS.

xlvii Orthogonal complement (正交補)

An orthogonal complement of a subspace W of a vector space V equipped with a bilinear form B is the set $W^\perp$ of all vectors in V that are orthogonal to every vector in W, of which $v \in V$ is orthogonal to $w \in V$ if and only if $B(v, w) = 0$.

xlviii Affine combination

An affine combination is a linear combination of points where all coefficients sum up to 1.

xlix Convex combination

A convex combination is a linear combination of points where all coefficients are non-negative and sum up to 1.

l Coordinate system (座標/坐標系統)

A coordinate system is a system that uses one or more numbers, or coordinates, to uniquely determine the points on a manifold.

li Projection (投影)

A (vector) projection onto a subset U of a vector space V is a linear operator $P : V \to U$ such that $P^2 = P$, denoted as $\text{proj}_U P$. The property $P^2 = P$ is called idempotency (冪等性). Its norm is called scalar projection denoted as $\text{comp}_U P$.

lii Orthographic projection (正射影)

An orthographic on a subset U of a vector space V is a projection $P : V \to U$ such that

$$\forall v \in V : v - P(v) \in U^*$$

where U^* is the orthogonal complement of U.

liii Hilbert space (希爾伯特空間)

A Hilbert space is a real or complex inner product space that is also a complete metric space with respect to the distance function induced by the inner product.

liv Angle (夾角) between two vectors

Given two vectors u, v in an inner product space, the angle θ between than is defined as

$$\theta = \arccos(u \cdot v).$$

lv Principal angles (主夾角) between two subspaces

Given two subspaces U, W of an inner product space with $\dim(U) = k \leq \dim(W) = m$, there exists then a sequence of k angles $0 \leq \theta_1 \leq \theta_2 \leq \ldots \leq \theta_k \leq \frac{\pi}{2}$ called the principal angles between U and W, the first one defined as

$$\theta_1 := \min \left\{ \arccos \left(\frac{|\langle u, w \rangle|}{\|u\| \|w\|} \right) \middle| u \in U \wedge w \in W \right\} = \angle(u_1 \in U, w_1 \in W),$$

The vectors u_1 and w_1 are the corresponding principal vectors.

The other principal angles and vectors are then defined recursively via

$$\theta_i := \min \left\{ \arccos \left(\frac{|\langle u, w \rangle|}{\|u\| \|w\|} \right) \middle| u \in U \wedge w \in W \wedge \forall j \in \mathbb{N} \wedge j < i : \langle u_i, u_j \rangle = \langle w_i, w_j \rangle = 0 \right\} = \angle(u_i \in U, w_i \in W$$

lvi Flat, hyperplane, or affine subspace (仿射子空間)

A flat or an affine subspace of an affine space is a subspace of it that is an affine space.

lvii Hypersurface

A hypersurface is a $(n-1)$-dimensional manifold that is a subspace of an n-dimensional manifold.

lviii Parallel (平行)

Two nonzero vectors u, v in a vector space over field F are called to be parallel if there exists a scalar $x \in F$ such that $xu = v$, denoted as $u \parallel v$.

Two affine subspaces are called to be parallel if the direction spaces of them D, E satisfies:

$$D \subseteq E \vee E \subseteq D,$$

denoted as $D \parallel E$.

lix Coplanar

Multiple elements of a vector or affine space are called coplanar if they are in the same 2D affine subspace.

lx Elementwise multiplication

For two vectors consists of scalar components $\mathbf{u} = (u_1, \dots, u_n)$ and $\mathbf{v} = (v_1, \dots, v_n)$, their elementwise multiplication, denoted as $\mathbf{u} \odot \mathbf{v}$, is defined as

$$\mathbf{u} \odot \mathbf{v} = (u_1 v_1, \dots, u_n v_n).$$

lxi Unit vector (單位向量) or normalized vector

A unit vector or a normalized vector in a normed vector space is a vector of norm 1, denoted by a lowercase letter with a circumflex or hat, as in $\hat{\mathbf{v}}$ (pronounced "v-hat"). The unit vector or normalized vector $\hat{\mathbf{u}}$ of a nonzero vector $\mathbf{u}$ in a normed vector space equipped with a norm $\|\cdot\|$ is defined as:

$$\hat{\mathbf{u}} = \frac{\mathbf{u}}{\|\mathbf{u}\|}.$$

lxii Affine independence

For an affine space U associated with vector space V, $u_0, u_1, \dots, u_n \in U$ are affinely independent iff $(u_1 - u_0), (u_2 - u_0), \dots, (u_n - u_0)$ are linearly independent in V.

lxiii Simplex

A k-simplex or k-dimensional simplex in an affine space U is a set

$$\{\sum_{i=0}^{k} \theta_i u_i \mid \sum_{i=0}^{k} \theta_i = 1 \wedge \theta_0, \theta_1, \dots, \theta_k \in \mathbb{R}_{\geq 0}\},$$

where

$$u_0, u_1, \dots, u_k \in U$$

are affine independent and called vertices of the simplex.

lxiv Gram matrix, Gramian matrix, or Gramian

For n vectors $v_1, v_2, \dots, v_n$ in an inner product space, the Gram matrix, Gramian matrix, or Gramian G is defined as an $n \times n$ matrix where

$$G_{ij} = \langle v_i, v_j \rangle.$$

The Gram determinant or Gramian is the determinant of the Gram matrix.

lxv Support of functional

For a function $f : U \subseteq \mathbb{R}^n \to \mathbb{R}$ (or $\mathbb{C}$), the support of it, denoted as $\operatorname{supp}(f)$, is defined as

$$\operatorname{supp}(f) = \complement(\{x \in U \mid f(x) \neq 0\}).$$

If $\operatorname{supp}(f)$ is a compact set, f is called with compact support. If $\operatorname{supp}(f)$ is a compact subset of a set $V \subseteq U$, f is called with compact support on V.

II Euclidean vector space

i Euclidean vector space

A Euclidean vector space is a finite-dimensional inner product space over the real numbers, denoted as $\mathbb{R}^n$ for dimension n. The norm-induced metric $d(v, w) = \|v - w\|$ is called the Euclidean distance

ii Standard (unit/basis) vector

For a Euclidean vector space, the standard (unit/basis) vectors are the unit vectors with one component being 1 and others being 0.

iii Euclidean affine space (歐幾里德仿射空間) or Euclidean space (歐幾里得空間)

A Euclidean space, also known as Euclidean affine space, is an affine space over $\mathbb{R}$ such that the associated vector space is a Euclidean vector space, and is equipped with the same norm and metric as the associated Euclidean vector space.

iv Affine hull (仿射包)

The affine hull $\operatorname{aff}(S)$ of a set of points S in a Euclidean space is the smallest affine space that contains all the points in S, namely, the set of all affine combinations of the points in S, i.e.:

$$\operatorname{aff}(S) = \left\{ \sum_{i=1}^{k} \alpha_i x_i : k > 0, x_i \in S, \alpha_i \in \mathbb{R}, \sum_{i=1}^{k} \alpha_i = 1 \right\}$$

v Convex hull (凸包)

The convex hull $\text{conv}(S)$ of a set of points S in a Euclidean space is the smallest convex set that contains all the points in S, namely, the set of all convex combinations of points in S, i.e.:

$$\text{conv}(S) = \left\{ \sum_{i=1}^{k} \alpha_i x_i : k > 0, x_i \in S, \alpha_i \in [0,1], \sum_{i=1}^{k} \alpha_i = 1 \right\}$$

vi Cartesian coordinate system (笛卡爾座標系統) or rectangular coordinate system (直角座標系統)

The Cartesian or rectangular coordinate system is a coordinate system in a Euclidean vector space E determined by an orthogonal basis of unit vectors such that the inner product of two vectors in E equals to the dot product of them, and that the norm of E is 2-norm.

vii Lattice points (格子點)

In a Cartesian coordinate system, vectors whose components are all integers are called lattice points.

viii Curvilinear coordinate system

A curvilinear coordinate system is a coordinate system for Euclidean space such that there exists an invertible map from a Cartesian coordinate system to it.

ix Standard, unit, or probability simplex

The standard, unit, or probability k-simplex is the simplex in $\mathbb{R}^{k+1}$ whose vertices are the unit vectors of each axis, that is, the set

$$\{(p_0, p_1, \dots, p_k) \mid \sum_{i=0}^{k} p_i = 1 \wedge p_0, p_1, \dots, p_k \in \mathbb{R}_{\geq 0}\}.$$

x Rectangle

An open rectangle in $\mathbb{R}^n$ is the product of n open intervals.

An closed rectangle in $\mathbb{R}^n$ is the product of n closed intervals.

Rectangle may refer to either, or product of n intervals, or product of n intervals satisfying certain constraints, depending on context, and often the case they make no non-trivial difference if not specified.

III Tensor

i Tensor

Let V be a vector space and V^* be its dual. A tensor of type (r, s) is a multilinear map

$$T : (V^*)^r \times V^s \to \mathbb{R},$$

where r is the number of contravariant slots, which take covectors in V^*, and s is the number of covariant slots, which take vectors in V. The order or rank of it is $r + s$.

This map

$$T : (V^*)^r \times V^s \to \mathbb{R}$$

is also a map

$$T : (V^*)^q \times V^t \to \{S : (S : (V^*)^{r-q} \times V^{s-t} \to \mathbb{R})\}$$

for any $q \in \mathbb{Z} \cap [0, r]$ and $t \in \mathbb{Z} \cap [0, s]$.

A scalar is a tensor of type $(0, 0)$. A vector is a tensor of type $(1, 0)$, which maps a covector to a real number, thus also called a contravariant tensor. A covector is a tensor of type $(0, 1)$, which maps a vector to a real number, thus also called a covariant tensor.. A matrix as a linear map $V \to V$ is a tensor of type $(1, 1)$. A matrix as a bilinear form $V \times V \to \mathbb{R}$ is a tensor of type $(0, 2)$. A matrix as $V^* \times V^* \to \mathbb{R}$ is a tensor of type $\mathbb{R}$.

It's conventional that a vector is a column vector and a covector is a row vector. Tensor convention follows.

A tensor can be represented as an array whose dimension is its rank when a basis is chosen.

ii Upper and lower indices

Dummy indices such as μ and ν are used to indicate contravariant and covariant vectors and tensors.

We use a superscript of r dummy indices and a subscript of s dummy indices to indicate a tensor of type (r, s), i.e., $(V^*)^r \times V^s \to \mathbb{R}$ for vector space V, such as $T^{\mu_1 \cdots \mu_r}_{\nu_1 \cdots \nu_s}$. Each ith component of each upper index μ is denoted by replacing μ with i. Each ith component of each lower index μ is denoted by replacing μ with i. For vectors, that is, $A^\mu = (A^0, A^1, A^2, A^3)$. For covectors, that is, $A_\mu = (A_0, A_1, A_2, A_3)$.

Each upper index μ is fixed to a summation $\sum_{\mu=0}^{\dim(V)}$ when it takes an input of the same upper index μ, i.e.

$$T^{\mu \cdots 1}_{\cdots 2} U^{\mu \cdots 3}_{\cdots 4} = \sum_{\mu=0}^{\dim(V)} T^{\mu \cdots 1}_{\cdots 2} U^{\mu \cdots 3}_{\cdots 4}.$$

Each lower index μ is fixed to a summation $\sum_{\mu=0}^{\dim(V)}$ when it takes an input of the same lower index μ, i.e.

$$T^{\cdots 1}_{\mu \cdots 2} U^{\cdots 3}_{\mu \cdots 4} = \sum_{\mu=0}^{\dim(V)} T^{\cdots 1}_{\mu \cdots 2} U^{\cdots 3}_{\mu \cdots 4}.$$

The indexing replaces the order of them in a formula and makes the multiplication notation of tensors work without extra notations.

IV Two dimensional Euclidean vector space

i Quadrant (象限)

I(+, +), II(-, +), III(-, -), IV(+, -).

V Three dimensional Euclidean vector space

i Right-hand rule (右手定則)

Right-hand rule is a common used orientation for a three-dimensional space. In a right-handed Cartesian coordinate system, if you point the thumb of your right hand in the positive x-axis direction and your index finger in the positive y-axis direction, then your middle finger (extended perpendicularly from the palm) will point in the positive z-axis direction. If not specified otherwise, right-handed Cartesian coordinate system is usually used.

ii Octant (卦限)

I(+, +, +), II(-, +, +), III(-, -, +), IV(+, -, +), V(+, +, -), VI(-, +, -), VII(-, -, -), VIII(+, -, -).

iii Cross product (叉積), external product (外積), or vector product (向量積)

The cross product of vector $\mathbf{a} = (a_1, a_2, a_3)$ and $\mathbf{b} = (b_1, b_2, b_3)$ is defined as:

$$\mathbf{a} \times \mathbf{b} = \det \begin{pmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{pmatrix},$$

where $\mathbf{i} = (1, 0, 0)$, $\mathbf{j} = (0, 1, 0)$, $\mathbf{k} = (0, 0, 1)$.

iv Basic properties of cross product

- Self cross product is the zero vector:

$$\mathbf{a} \times \mathbf{a} = \mathbf{0}.$$

- Anticommutative law (反交換律):

$$\mathbf{a} \times \mathbf{b} = -\mathbf{b} \times \mathbf{a}.$$

- Left and right distributive over addition and subtraction:

$$\mathbf{a} \times (\mathbf{b} \pm \mathbf{c}) = \mathbf{a} \times \mathbf{b} \pm \mathbf{a} \times \mathbf{c}.$$

$$(\mathbf{b} \pm \mathbf{c}) \times \mathbf{a} = \mathbf{b} \times \mathbf{a} \pm \mathbf{c} \times \mathbf{a}.$$

- Commutative and associative over scalar multiplication:

$$(r\mathbf{a}) \times \mathbf{b} = a \times (r\mathbf{b}) = r(\mathbf{a} \times \mathbf{b}).$$

- Two vector in $\mathbb{R}^3$ are parallel iff their cross product is the zero vector.

v Triple product identity (三重積恆等式)/Jacobi identity (雅可比恆等式)

$$\mathbf{A}\times(\mathbf{B}\times\mathbf{C})+\mathbf{B}\times(\mathbf{C}\times\mathbf{A})+\mathbf{C}\times(\mathbf{A}\times\mathbf{B})=0.$$

$$\mathbf{A}\cdot(\mathbf{B}\times\mathbf{C})=\mathbf{B}\cdot(\mathbf{C}\times\mathbf{A})=\mathbf{C}\cdot(\mathbf{A}\times\mathbf{B})=\begin{vmatrix}\mathbf{A}\\\mathbf{B}\\\mathbf{C}\end{vmatrix}$$

$$(\mathbf{A}\times\mathbf{B})\times\mathbf{C}=(\mathbf{A}\cdot\mathbf{C})\mathbf{B}-(\mathbf{B}\cdot\mathbf{C})\mathbf{A}.$$

vi Scalar triple product

$$[\mathbf{a},\mathbf{b},\mathbf{c}]=\det\big(\mathbf{a},\mathbf{b},\mathbf{c}\big)=\mathbf{a}\cdot(\mathbf{b}\times\mathbf{c})$$

2 Matrix (矩陣)

I Matrix

A matrix (plural: matrices) is a rectangular array of numbers (or other mathematical objects), called the entries (元或元素) of the matrix. Most commonly, a matrix over a field F is a rectangular array of elements of F.

The size of a matrix is defined by the number of rows (zh-hant: 列/zh-hans: 行) and columns (zh-hant: 行/zh-hans: 列) it contains. A matrix with m rows and n columns is called an $m\times n$ matrix, or m-by-n matrix, where m and n are called its dimensions (階). A matrix with the same number of rows and columns n is called a n square matrix (方陣). A matrix with an infinite number of rows or/and columns is called an infinite matrix. A matrix with exactly one row is called a row vector. A matrix with exactly one column is called a column vector.

The entry in the i-th row and j-th column, called the i,jth entry (第 i,j 元) or ijth entry (第 ij 元), is commonly denoted as $a_{i,j}$ or a_{ij}.

An $m\times n$ matrix $\mathbf{A}$ is commonly denoted as:

$$\mathbf{A}=\begin{bmatrix}a_{11}&a_{12}&\dots&a_{1n}\\a_{21}&a_{22}&\dots&a_{2n}\\\vdots&\vdots&\ddots&\vdots\\a_{m1}&a_{m2}&\dots&a_{mn}\end{bmatrix},$$

$$\mathbf{A}=\begin{pmatrix}a_{11}&a_{12}&\dots&a_{1n}\\a_{21}&a_{22}&\dots&a_{2n}\\\vdots&\vdots&\ddots&\vdots\\a_{m1}&a_{m2}&\dots&a_{mn}\end{pmatrix},$$

$$\mathbf{A}=[a_{ij}]_{1\le i\le m,1\le j\le n},$$

$$\mathbf{A}=(a_{ij})_{1\le i\le m,1\le j\le n},$$

$$\mathbf{A}=[a_{ij}]_{m\times n},$$

or

$$\mathbf{A}=(a_{ij})_{m\times n}.$$

The i, ith entries are called diagonal entries (對角線元).

A matrix with only one row is called a row matrix or row vector, and a matrix with only one column is called a column matrix or column vector.

The ith row of the matrix **A** is commonly denoted as $\mathbf{A}_{i,:}$, and the jth column is commonly denoted as $\mathbf{A}_{:,j}$.

The set of all $m \times n$ matrices over $\mathbb{F}$ is denoted as $\mathbb{F}^{m \times n}$.

II Basic operations

i Submatrix

A submatrix of a matrix M is obtained by deleting from M entire rows, entire columns, or both.

ii Matrix equality

We say that two matrices $\mathbf{A} = [a_{ij}]_{m \times n}$ and $\mathbf{B} = [b_{ij}]_{m \times n}$ are equal if

$$\forall 1 \leq i \leq m, 1 \leq j \leq n : \ a_{ij} = b_{ij}.$$

iii Matrix addition

The sum of two $m \times n$ matrices **A** and **B** is calculated entrywise.

iv Scalar multiplication

The product $c\mathbf{A}$ of a scalar c is calculated by multiplying every entry of **A** by c.

v Matrix multiplication (矩陣乘法)

Multiplication of two matrices is defined only if the number of columns of the left matrix is the same as the number of rows of the right matrix. The matrix product **AB** of an $m \times n$ matrix **A** and $n \times p$ matrix **B** is the $m \times p$ matrix whose ijth entry is the dot product of the ith row of **A** and the jth column of **B**.

vi Positive integer exponent

Given a square matrix **A** and a positive integer n, $\mathbf{A}^n$ is defined as $\prod_{i=1}^{n} \mathbf{A}$.

vii Zero matrix (零矩陣)

A zero matrix **O** over a field F is a matrix of which all entries are the additive identity 0_F of F.

viii Identity matrix (單位矩陣)

An identity matrix **I** over a field F is a square matrix of which all diagonal entries are the multiplicative identity 1_F of F and all off-diagonal entries are the multiplicative absorbing element 0_F of F (i.e. $\forall x \in F : \ x \times 0_F = 0_F$).

ix Hadamard product (哈達瑪乘積), Schur product (舒爾乘積), elementwise product, or entrywise product (逐項乘積)

For two $m \times n$ matrices **A** and **B**, the Hadamard product $\mathbf{A} \odot \mathbf{B}$ or $\mathbf{A} \circ \mathbf{B}$ is an $m \times n$ matrix defined as:

$$(\mathbf{A} \odot \mathbf{B})_{ij} = (\mathbf{A})_{ij}(\mathbf{B})_{ij}.$$

x Kronecker product (克羅內克積)

If **A** is an $m \times n$ matrix and **B** is a $p \times q$ matrix, then the Kronecker product $\mathbf{A} \otimes \mathbf{B}$ is the $pm \times qn$ block matrix:

$$(\mathbf{A} \otimes \mathbf{B})_{i,j} = a_{\left\lceil \frac{i}{p} \right\rceil, \left\lceil \frac{j}{q} \right\rceil} b_{((i-1) \mod p)+1, ((j-1) \mod q)+1}$$

xi Merging matrices

The matrix formed by merging two matrices **A** and **B** with the same number of rows left and right is denoted as:

$$\begin{bmatrix} \mathbf{A} & \mathbf{B} \end{bmatrix}$$

or

$$\begin{pmatrix} \mathbf{A} & \mathbf{B} \end{pmatrix}$$

The matrix formed by merging two matrices **A** and **B** with the same number of columns up and down is denoted as:

$$\begin{bmatrix} \mathbf{A} \\ \mathbf{B} \end{bmatrix}$$

or

$$\begin{pmatrix} \mathbf{A} \\ \mathbf{B} \end{pmatrix}$$

xii Property

For any scalars r and s, $m \times n$ matrices A, B, C, D, zero matrix O, and identity matrix I, the following statements are true,

$$A + B = B + A$$

$$A + (B + C) = (A + B) + C$$

$$r(A + B) = rA + rB$$

$$rA + sA = (r + s)A$$

$$(rs)A = r(sA) = s(rA)$$

$$(AB)C = A(BC)$$

$$r(AB) = (rA)B = A(rB)$$

$$A(B + C) = AB + AC$$

$$(A + B)C = AC + BC$$

$$(A+B)(C+D) = AC + AD + BC + BD$$

$$A \odot B = B \odot A$$

$$A \odot (B \odot C) = (A \odot B) \odot C$$

$$A \odot (B + C) = A \odot B + A \odot C$$

$$IA = A$$

$$AI = A$$

$$A + O = A$$

$$AO = O$$

$$OA = O$$

the following statements are not necessarily true,

$$AB = BA$$

Proof. Counterexample:

$$A = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$$

$$B = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

$$AB = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$$

$$BA = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$$

□

$$(AB = AC \wedge A \neq O) \Rightarrow B = C$$

Proof. Counterexample:

$$A = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$$

$$B = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$C = \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}$$

$$AB = AC = A$$

□

$$(ACB = ADB \wedge A \neq O \wedge B \neq O) \Rightarrow C = D$$

Proof. Counterexample:

$$A = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$$

$$B = C = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$D = \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}$$

$$ACB = AC = ADB = AD = A$$

□

$$A^2 = O \Rightarrow A = O$$

Proof. Counterexample:

$$A = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$$

□

$$AB = O \Rightarrow (A = O \vee B = O)$$

Proof. Counterexample:

$$A = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$$

$$B = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$$

□

$$AB = B \Rightarrow A = I$$

Proof. Counterexample:

$$A = B = AB = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$$

□

xiii Zero exponent

Given an $n \times n$ matrix **A** and $n \times n$ identity matrix **I**, $\mathbf{A}^0$ is defined as **I**.

III Matrix row and column operations

i Row operations (zh-hant: 列/zh-hans: 行運算) and Column operations (zh-hant: 行/zh-hans: 列運算)

(Elementary) row operations include:

- Row addition (zh-hant: 列/zh-hans: 行相加) operation: Adding a row to another.
- Row scaling/multiplication (zh-hant: 列/zh-hans: 行加倍) operation: Multiplying all entries of a row by a nonzero constant.
- Row swap/switching/interchange (zh-hant: 列/zh-hans: 行交換) operation: Interchanging two rows of a matrix.

(Elementary) column operations are the same with row replaced with column.

If one matrix can be transformed to another through a sequence of elementary row operations, the two matrices are called row-equivalent.

If one matrix can be transformed to another through a sequence of elementary column operations, the two matrices are called column-equivalent.

ii Zero row, nonzero row, and leading entry

A row whose all entries are zero is called a zero row, otherwise it is called a nonzero row. The left-most nonzero entry of a nonzero row is called leading entry.

iii Transpose (轉置)

The transpose of an $m \times n$ matrix $\mathbf{A}$ is the $n \times m$ matrix $\mathbf{A}^\top$:

$$(\mathbf{A}^\top)_{i,j} = a_{j,i}$$

Inverse map as linear map is transpose as matrix.

iv Symmetric matrix (對稱矩陣)

$\mathbf{A}$ is called a symmetric matrix iff $\mathbf{A} = \mathbf{A}^\top$.

v Skew-symmetric matrix/antisymmetric matrix (反對稱矩陣)

$\mathbf{A}$ is called a skew-symmetric matrix or an antisymmetric matrix if $\mathbf{A} = -\mathbf{A}^\top$.

vi Transpose identities

For an index set of matrices $\mathbf{A} : [1, n] \cap \mathbb{N} \to \mathbb{F}^{p \times q};\ i \mapsto \mathbf{A}(i)$ for some fixed p, q,

$$\left(\sum_{i=1}^{n} \mathbf{A}(i) \right)^\top = \sum_{i=1}^{n} (\mathbf{A}(i))^\top$$

Proof.

$$\left(\sum_{i=1}^{n}\mathbf{A}(i)\right)_{jk}=\sum_{i=1}^{n}\mathbf{A}(i)_{jk}$$

$$\left(\left(\sum_{i=1}^{n}\mathbf{A}(i)\right)^{\top}\right)_{jk}=\sum_{i=1}^{n}\mathbf{A}(i)_{kj}$$

$$=\sum_{i=1}^{n}\left((\mathbf{A}(i))^{\top}\right)_{jk}$$

$$=\left(\sum_{i=1}^{n}(\mathbf{A}(i))^{\top}\right)_{jk}$$

□

For two matrices $X\in\mathbb{F}^{p\times q},Y\in\mathbb{F}^{q\times r}$,

$$(XY)^{\top}=Y^{\top}X^{\top}$$

Proof.

$$(XY)_{jk}=\sum_{i=1}^{q}X_{ji}Y_{ik}$$

$$((XY)^{\top})_{jk}=\sum_{i=1}^{q}X_{ki}Y_{ij}$$

$$=\sum_{i=1}^{q}(X^{\top})_{ik}(Y^{\top})_{ji}$$

$$=(Y^{\top}X^{\top})_{jk}$$

□

For an index set of matrices $\mathbf{A}:\ [1,n]\cap\mathbb{N}\to\mathbb{F}^{\mathbb{N}\times\mathbb{N}};\ i\mapsto\mathbf{A}(i)$ such that $\mathbf{A}(i)\mathbf{A}(i+1)$ is defined for all $i\in[1,n)\cap\mathbb{N}$,

$$\left(\prod_{i=1}^{n}\mathbf{A}(i)\right)^{\top}=\prod_{i=1}^{n}(\mathbf{A}(n-i+1))^{\top}$$

Proof. Prove by mathematical induction on n.

Base case $n=1$:

$$\left(\prod_{i=1}^{1}\mathbf{A}(i)\right)^{\top}=\prod_{i=1}^{1}(\mathbf{A}(1-i+1))^{\top},$$

which is true.

Assume the statement holds for $n=k$:

$$\left(\prod_{i=1}^{k}\mathbf{A}(i)\right)^{\top}=\prod_{i=1}^{k}(\mathbf{A}(k-i+1))^{\top}$$

For $n = k + 1$, we have

$$\begin{aligned}\left(\prod_{i=1}^{k+1}\mathbf{A}(i)\right)^{\top} &= \left(\prod_{i=1}^{k}\mathbf{A}(i)A(k+1)\right)^{\top}\\ &= (A(k+1))^{\top}\left(\prod_{i=1}^{k}\mathbf{A}(i)\right)^{\top}\\ &= (A(k+1))^{\top}\prod_{i=1}^{k}(\mathbf{A}(k-i+1))^{\top}\\ &= \prod_{i=1}^{k+1}(\mathbf{A}(k-i+1))^{\top}\end{aligned}$$

□

vii Conjugate transpose (共軛轉置)/Hermitian conjugate (埃爾米特共軛)/Hermitian transpose (埃爾米特轉置)

The conjugate transpose of an $m \times n$ matrix $\mathbf{A}$ is the $n \times m$ matrix $\mathbf{A}*$:

$$(\mathbf{A}^*)_{ij} = \overline{a_{ji}}$$

viii Conjugate transpose identities

For an index set of matrices $\mathbf{A} : [1, n] \cap \mathbb{N} \to \mathbb{C}^{p\times q};\ i \mapsto \mathbf{A}(i)$ for some fixed p, q,

$$\left(\sum_{i=1}^{n}\mathbf{A}(i)\right)^{*} = \sum_{i=1}^{n}(\mathbf{A}(i))^{*}$$

For two matrices $X \in \mathbb{C}^{p\times q}, Y \in \mathbb{C}^{q\times r}$,

$$(XY)^* = Y^*X^*$$

For an index set of matrices $\mathbf{A} : [1, n] \cap \mathbb{N} \to \mathbb{C}^{\mathbb{N}\times\mathbb{N}};\ i \mapsto \mathbf{A}(i)$ such that $\mathbf{A}(i)\mathbf{A}(i+1)$ is defined for all $i \in [1, n) \cap \mathbb{N}$,

$$\left(\prod_{i=1}^{n}\mathbf{A}(i)\right)^{*} = \prod_{i=1}^{n}(\mathbf{A}(n-i+1))^{*}$$

IV Trace (跡)

The trace of a matrix $\mathbf{A}$ is the sum of all diagonal entries of $\mathbf{A}$, denoted as $\mathrm{tr}(\mathbf{A})$.

V Cofactor matrix, adjugate matrix, determinant, and inverse

Let $\mathbf{M}_{ij}$ of $\mathbf{A}$ be the determinant of the submatrix of $\mathbf{A}$ without the ith row and jth column.

i Determinant (行列式或判別式)

The determinant of a 1×1 matrix is the only element of it; the determinant of an $n \times n$ matrix $\mathbf{A}$, denoted $\det(\mathbf{A})$ or

$$\begin{vmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \dots & a_{nn} \end{vmatrix},$$

is

$$\det(\mathbf{A}) = \sum_{i=1}^{n} a_{ij}(-1)^{i+j}\mathbf{M}_{ij} \quad j \leq n \in \mathbb{N}$$

$$\det(\mathbf{A}) = \sum_{j=1}^{n} a_{ij}(-1)^{i+j}\mathbf{M}_{ij} \quad i \leq n \in \mathbb{N}$$

ii Determinant identities

Let $\mathbf{A}$ and $\mathbf{A}_i$ be square matrices.

$$\det\left(\sum_{i=1}^{n} \mathbf{A}_i\right) = \sum_{i=1}^{n} \det\left(\mathbf{A}_i\right)$$

$$\det\left(\prod_{i=1}^{n} \mathbf{A}_i\right) = \prod_{i=1}^{n} \det\left(\mathbf{A}_i\right)$$

$$\det(\mathbf{A}) \neq 0 \implies \det\left(\mathbf{A}^n\right) \neq 0$$

iii Cofactor matrix (餘因子矩陣)

The cofactor matrix of an $n \times n$ matrix $\mathbf{A}$ is the $n \times n$ matrix $\operatorname{cof}(\mathbf{A})$:

$$[\operatorname{cof}(\mathbf{A})]_{ij} = \left((-1)^{i+j}\mathbf{M}_{ij}\right)$$

iv Adjugate matrix (伴隨矩陣)/classical adjoint (經典伴隨矩陣)

The adjugate matrix or classical adjoint matrix of an $n \times n$ matrix $\mathbf{A}$ is the $n \times n$ matrix $\operatorname{adj}(\mathbf{A})$:

$$\operatorname{adj}(\mathbf{A}) = \operatorname{cof}(\mathbf{A})^{\mathsf{T}}$$

v Inverse (反方陣、逆方陣、反矩陣或逆矩陣), multiplicative inverse (乘法反方陣), or reciprocal

The inverse of an $n \times n$ matrix $\mathbf{A}$ is an $n \times n$ matrix $\mathbf{A}^{-1}$ such that:

$$\mathbf{A}\mathbf{A}^{-1} = \mathbf{A}^{-1}\mathbf{A} = \mathbf{I}$$

vi Negative integer exponent

Given an invertible matrix $\mathbf{A}$ and a negative integer n, $\mathbf{A}^n$ is defined as $(\mathbf{A}^{-1})^n$.

vii Invertible matrix (可逆矩陣), orthogonal matrix (正交矩陣), orthonormal matrix (正則矩陣), non-singular matrix (非奇異矩陣), or non-degenerate matrix (非退化矩陣)

An invertible matrix, orthogonal matrix, orthonormal matrix, non-singular matrix, or non-degenerate matrix is a matrix such that its inverse exists.

Let $\mathbf{A}$ be an $n \times n$ matrix, the following prepositions are equivalent to "$\mathbf{A}$ is an invertible matrix":

$$\exists \mathbf{A}^{-1}$$
$$\exists \left(\mathbf{A}^{\top}\right)^{-1}$$
$$\exists \left(\mathbf{A}^{\top}\mathbf{A}\right)^{-1}$$
$$\operatorname{rank}(\mathbf{A}) = n$$
$\mathbf{A}$存在非零的特徵值

viii Non-invertible matrix (不可逆矩陣), non-orthogonal matrix (非正交矩陣), non-orthonormal matrix (非正則矩陣), singular matrix (奇異矩陣), or degenerate matrix (退化矩陣)

An non-invertible matrix, non-orthogonal matrix, non-orthonormal matrix, singular matrix, or degenerate matrix is a matrix that is not an invertible matrix.

ix Inverse of a two-by-two matrix

For an invertible 2×2 matrix $\mathbf{A}$:

$$\mathbf{A} = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \implies \mathbf{A}^{-1} = \frac{1}{\det(\mathbf{A})} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

x Computing inverse with row operations

Let $\mathbf{A}$ be an $n \times n$ matrix and $\mathbf{I}$ be the $n \times n$ identity matrix. If $\begin{pmatrix} \mathbf{A} & \mathbf{I} \end{pmatrix}$ can become $\begin{pmatrix} \mathbf{I} & \mathbf{B} \end{pmatrix}$ after row operations, then

$$\mathbf{B} = \mathbf{A}^{-1}$$

xi Unitary matrix (么正矩陣或酉矩陣)

An unitary matrix is a square matrix whose inverse equals to its conjugate transpose.

xii Determinant property

- When a row (or column) is added to another, the determinant remains unchanged.
- When all entries of a row (or column) is multiplied by a real number n, the determinant is multiplied by n.
- When two rows (or columns) are swapped, the determinant is multiplied by -1.
- Given matrix $\mathbf{A}$ and its inverse $\mathbf{A}^{-1}$, $\det\left(\mathbf{A}^{-1}\right) = \frac{1}{\det(\mathbf{A})}$.

xiii Inverse identities

Let $\mathbf{A}$ and $\mathbf{A}_i$ be invertible matrices, $\mathbf{B}$ be a matrix of the same size of $\mathbf{A}$.

$$\mathbf{A}^{-1} = \frac{\operatorname{adj}(\mathbf{A})}{\det(\mathbf{A})}$$

$$\left(\mathbf{A}^\top\right)^{-1} = \left(\mathbf{A}^{-1}\right)^\top$$

$$\left(\prod_{i=1}^{n} \mathbf{A}_i\right)^{-1} = \prod_{i=1}^{n} \left(\mathbf{A}_{n-i+1}\right)^{-1}$$

$$(\mathbf{A}^{-1}\mathbf{B}\mathbf{A})^n = \mathbf{A}^{-1}\mathbf{B}^n\mathbf{A}, \quad n \in \mathbb{N}$$

Let A, B be square matrix of same size,

$$AB = I \Rightarrow (\exists A^{-1} \wedge A^{-1} = B)$$

xiv Vandermonde matrix (范德蒙矩陣)

A Vandermonde matrix $\mathbf{V}$ is:

$$V_{i,j} = {\alpha_i}^{j-1}$$

An $n \times n$ Vandermonde matrix $\mathbf{V}$ satisfies:

$$\det(V) = \prod_{1 \le i < j \le n} (\alpha_j - \alpha_i)$$

VI Diagonal matrix (對角矩陣)

i Diagonal matrix

A diagonal matrix over a field F is a square matrix over F of which all non-diagonal entries are the additive identity of F.

ii Diagonal matrix identities

If:

$$\mathbf{D} = \begin{bmatrix} d_1 & 0 & \cdots & 0 \\ 0 & d_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & d_n \end{bmatrix}$$

where all $d_i \neq 0$, then:

$$\mathbf{D}^{-1} = \begin{bmatrix} \dfrac{1}{d_1} & 0 & \cdots & 0 \\ 0 & \dfrac{1}{d_2} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \dfrac{1}{d_n} \end{bmatrix}$$

Let $\mathbf{E}$, $\mathbf{F}$ be two $n \times n$ diagonal matrices:

$$\mathbf{EF} = [e_{ij} f_{ij}]_{1 \le i,j \le n}$$

VII System of Linear Equations (線性方程組)

i System of equations (方程組)

A solution to a system of equations is a choice of variables such that all equations are satisfied. Two systems of equations are called equivalent if they have exactly the same set of solutions.

- **Consistent system (相容方程組)**: A system of equations is consistent iff it has at least one solution.
- **Inconsistent/contradictory system (相容方程組)**: A system of equations is inconsistent or contradictory iff it has no solution.
- **Dependent system (相依方程組)**: A consistent system of equations is dependent iff it has infinitely many solutions. For a dependent system, we can solve some of the variables, called basic variables, in terms of some other variables, called free variables, such that the free variables are independent from each other and that a choice of free variables uniquely determines basic variables, providing a general solution of the system.
- **Independent system (獨立方程組)**: A system of equations is independent iff it has finitely many solutions.

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ii Linear Equation (線性方程)

A linear equation is an equation that may be put in the from:

$$\sum_{k=1}^{n} a_k x_k = c,$$

where a_k are called coefficients and c is called constant term.

If $c = 0$, the linear equation is called a homogeneous linear equation; otherwise, it is called a heterogeneous or non-homogeneous linear equation.

iii System of Linear Equations (線性方程組) or linear system

A system of linear equations is a collection of two or more linear equations involving the same tuple of variables.

iv Independent system of linear equations has exactly one solution

An independent system of linear equations must have exactly one solution.

Proof. If more than one solution exist, their linear combinations form infinitely many solutions. □

v Coefficient matrix (係數矩陣)

The coefficients of a system of linear equations written as a matrix, with one row for each equation.

vi Constant term vector

The column vector of the constant terms on the right side of the equal signs of a system of linear equations.

vii Augmented matrix (增廣矩陣)

Coefficient matrix with the constant term vector augmented to the last column of the matrix.

viii Row meanings

- Two systems of linear equations have the same set of solutions iff their augmented matrices are row-equivalent.
- Whenever an augmented matrix contains a row in which the only nonzero entry lies in the last column, the corresponding system of linear equations has no solution.

ix Cramer Rule or Cramer formula (克拉瑪公式)

Given a system of linear equations that can be written in the form:

$$\mathbf{AX} = \mathbf{C}$$

where **A** is an $n \times n$ coefficient matrix, **X** is a variable matrix:

$$\mathbf{X} = \left(x_1, x_2, x_3, \dots, x_n\right)^\top,$$

and **C** is a coefficient matrix:

$$\mathbf{C} = \left(c_1, c_2, c_3, \dots, c_n\right)^\top.$$

Let $\mathbf{A}_i$ be **A** with the ith column replaced by **C**. The solution to the system of linear equations is given by:

$$\begin{cases} x_i = \dfrac{\det\left(\mathbf{A}_i\right)}{\det\left(\mathbf{A}\right)}, & \det\left(\mathbf{A}\right) \neq 0 \\ \text{inconsistent}, & \det\left(\mathbf{A}\right) = 0 \wedge \bigvee_{i=1}^{n} \det\left(\mathbf{A}_i\right) \neq 0 \\ \text{dependent}, & \det\left(\mathbf{A}\right) = 0 \wedge \bigvee_{i=1}^{n} \det\left(\mathbf{A}_i\right) = 0 \end{cases}$$

where $x_i = \dfrac{\det\left(\mathbf{A}_i\right)}{\det\left(\mathbf{A}\right)}$ can be geometrically viewed as the volume of the parallelogram formed by the column vectors of $\mathbf{A}_i$ divided by the volume of the parallelogram formed by the column vectors of **A**.

Particularly, when **C** is a zero matrix, that is,

$$\mathbf{AX} = \mathbf{0},$$

the solution to the system of linear equations is given by:

$$\begin{cases} x_i = 0, & \det\left(\mathbf{A}\right) \neq 0 \\ \text{dependent}, & \det\left(\mathbf{A}\right) = 0 \end{cases}$$

VIII Row echelon form

i (General) row echelon form (REF) (（廣義）zh-hant: 列/zh-hans: 行階梯形式)

A matrix is said to be in row echelon form iff:

1. Each nonzero lies above every zero row.
2. The leading entry of every nonzero row lies in a column to the right of the column containing the leading entry of any row above.

Thus, for a leading entry of any row, all entries below it in the same column are zero.

ii Reduced row echelon form (RREF)

A matrix is said to be in reduced row echelon form iff:

1. It is in row echelon form.
2. Each leading entry is the only nonzero entry in the column.
3. The leading entries of all nonzero rows are 1.

iii Pivot position

For a matrix A and its (unique, which will be proven later) reduced row echelon form R, the position containing a leading entry of a nonzero row is called pivot position, a column containing a pivot position is called pivot column, and a row containing a pivot position is called pivot row.

iv Gaussian Elimination (高斯消去法) or row reduction

Gaussian elimination is a recursive algorithm to solve a system of linear equations by finding the row-equivalent row echelon form of the augmented matrix.

1. Determine the leftmost nonzero column. This is a pivot column, and the topmost position in this column is a pivot position.
2. In the pivot column, choose any nonzero entry, and perform the appropriate row interchange to bring this entry into the pivot position.
3. Add an appropriate multiple of the row containing the pivot position to each lower row in order to change each entry below the pivot position into 0.
4. Ignore the row containing the pivot position. If there is a nonzero row below it, repeat steps 1–4 (called forward pass) on the submatrix that remains.

v Gaussian-Jordan Elimination

Gaussian-Jordan elimination is an algorithm to solve a system of linear equations by finding the row-equivalent reduced row echelon form of the augmented matrix. First do Gaussian

elimination to obtain a row-equivalent row echelon form, and then, from the topmost to bottommost nonzero row, do the following (called backward pass):

1. Perform the appropriate row scaling the make the leading entry, which is a pivot position, 1.
2. Add an appropriate multiple of the row to each upper row in order to change each entry above the pivot position into 0.

vi Row-equivalent reduced echelon form to general solution

Given the row-equivalent reduced row echelon form of the augmented matrix of a linear system, if it has no row with leading entry in the rightmost column, we can obtain the general solution by

1. Assign the variables of non-pivot columns as free variables and variables of pivot columns as basic variables.
2. For each nonzero row, write an equation of the variable of the leading entry as a linear combination of the variables (constant term for the rightmost column and free variables for others) of other nonzero entries.

vii Row-equivalent reduced echelon form test for consistency of linear system

A system of linear equations is consistent iff the reduced row echelon form of its augmented matrix has no row with leading entry in the rightmost column.

Proof. If the reduced row echelon form of its augmented matrix has row with leading entry in the rightmost column, it is obviously inconsistent. If the reduced row echelon form of its augmented matrix has no row with leading entry in the rightmost column, we can construct a general solution. □

viii Existence and uniqueness of row-equivalent reduced row echelon form

Every matrix can be transformed into one and only one matrix in reduced row echelon form by means of a sequence of elementary row operations.

Proof. TODO □

IX Subspaces of matrices and linear maps

i Four fundamental subspaces of a matrix (矩陣的四大子空間)

The four fundamental subspaces of a matrix are the row space, column space, null space, and left null space.

ii Column space (zh-hant: 行/zh-hans: 列空間)

The column space of **A**, denoted as Col(**A**) or **C**(**A**), is defined as the span of the set of column vectors of **A**.

iii Row space (zh-hant: 列/zh-hans: 行空間)

The row space of **A** is defined as the span of the set of the row vectors of **A**, that is, Col($\mathbf{A}^\top$).

iv Null space (零空間), kernel (核), or kernel space (核空間)

The kernel of a linear map $L: V \to W$, denoted as ker(L) or Null(L), is defined as:

$$\ker(L) = \{v \in V : L(v) = 0\}.$$

$\dim\big(\ker(\mathbf{A})\big)$ is called nullity (零化度).

v Left null space (左零空間), left kernel (左核), or left kernel space (左核空間)

The left null space of a linear map $L: V \to W$ is defined as $\ker\big(L^{-1}\big)$.

$\dim\big(\ker\big(L^{-1}\big)\big)$ is called left nullity.

vi Image space (像空間) or range (值域)

The image space of a linear map $T: V \to W$, denoted as im(T), is defined as,

$$\operatorname{im}(T) = \{T(v) : v \in V\}.$$

vii Cokernel (餘核)

The cokernel of a linear map $T: V \to W$, denoted as coker(T), is defined as,

$$\operatorname{coker}(T) = W \setminus \operatorname{im}(T).$$

viii Rank (秩)

The rank of a matrix **A**, denoted as rank(**A**) or rk(**A**), is defined as the dimension of its column space (or row space, which are the same by constant rank theorem), that is,

$$\operatorname{rank}(\mathbf{A}) = \dim\big(\operatorname{Col}(\mathbf{A})\big)$$

ix Constant rank theorem (常秩定理)

For any matrix **A**:

$$\dim\big(\operatorname{Col}(\mathbf{A})\big) = \dim\big(\operatorname{Col}(\mathbf{A}^\top)\big)$$

Proof. Let **A** be an $m \times n$ matrix, $r = \dim\big(\operatorname{Col}(\mathbf{A})\big)$, $c_1, c_2, \dots, c_r$ be a basis of Col(**A**), matrix $\mathbf{C} = [c_1, c_2, \dots, c_r]$, and $r \times n$ matrix **R** such that $\mathbf{A} = \mathbf{CR}$.

Since:

$$\operatorname{Col}(\mathbf{A}^\top) \subseteq \operatorname{Col}(\mathbf{R}^\top),$$

$$\dim\big(\text{Col}(\mathbf{A}^\top)\big) \leq \dim\big(\text{Col}(\mathbf{R}^\top)\big)$$

Because there are only r rows in $\mathbf{R}$,

$$\dim\big(\text{Col}(\mathbf{R}^\top)\big) \leq \dim\big(\text{Col}(\mathbf{A})\big)$$

Similarly, we can obtain that

$$\dim\big(\text{Col}(\mathbf{A})\big) \leq \dim\big(\text{Col}(\mathbf{R}^\top)\big) \leq \dim\big(\text{Col}(\mathbf{A}^\top)\big).$$

Therefore,

$$\dim\big(\text{Col}(\mathbf{A})\big) = \dim\big(\text{Col}(\mathbf{R}^\top)\big) = \dim\big(\text{Col}(\mathbf{A}^\top)\big).$$

□

x Rank-nullity theorem (秩—零化度定理)

For any $m \times n$ matrix $\mathbf{A}$:

$$\text{rank}(\mathbf{A}) + \dim\big(\ker(\mathbf{A})\big) = n$$

Proof. Let $\mathbf{A}$ be an $m \times n$ matrix with rank r. We will show that there exists a set of $n - r$ linearly independent solutions to the homogeneous system $\mathbf{Ax} = \mathbf{0}_m$ where $\mathbf{x}$ is an n-dimensional column vector.

To do this, we will produce an $n \times (n - r)$ matrix $\mathbf{X}$ whose columns form a basis of the null space of $\mathbf{A}$.

Without loss of generality, assume that the first r columns of $\mathbf{A}$ are linearly independent. So, we can write

$$\mathbf{A} = \begin{pmatrix} \mathbf{A}_1 & \mathbf{A}_2 \end{pmatrix},$$

where

- $\mathbf{A}_1$ is an $m \times r$ matrix with r linearly independent column vectors, and
- $\mathbf{A}_2$ is an $m \times (n - r)$ matrix such that each of its $n - r$ columns is a linear combination of the columns of $\mathbf{A}_1$.

This means that $\mathbf{A}_2 = \mathbf{A}_1\mathbf{B}$ for some $r \times (n - r)$ matrix $\mathbf{B}$, and hence,

$$\mathbf{A} = \begin{pmatrix} \mathbf{A}_1 & \mathbf{A}_1\mathbf{B} \end{pmatrix}.$$

Let

$$\mathbf{X} = \begin{pmatrix} -\mathbf{B} \\ \mathbf{I}_{n-r} \end{pmatrix},$$

Then,

$$\mathbf{AX} = \begin{pmatrix} \mathbf{A}_1 & \mathbf{A}_1\mathbf{B} \end{pmatrix} \begin{pmatrix} -\mathbf{B} \\ \mathbf{I}_{n-r} \end{pmatrix} = -\mathbf{A}_1\mathbf{B} + \mathbf{A}_1\mathbf{B} = \mathbf{0}_{m \times (n-r)}.$$

Therefore, each of the $n - r$ columns of $\mathbf{X}$ is a solution of $\mathbf{Ax} = \mathbf{0}_m$.

Furthermore, we claim that the $n-r$ columns of $\mathbf{X}$ are linearly independent. We show that $\mathbf{0}_{n-r}$ is the only $(n-r)$-dimensional column vector $\mathbf{u}$ such that

$$\mathbf{Xu} = \mathbf{0}_n$$

$$\begin{pmatrix} -\mathbf{B} \\ \mathbf{I}_{n-r} \end{pmatrix} \mathbf{u} = \begin{pmatrix} -\mathbf{Bu} \\ \mathbf{u} \end{pmatrix} = \begin{pmatrix} \mathbf{0}_r \\ \mathbf{0}_{n-r} \end{pmatrix}$$

We next prove that any solution of $\mathbf{Ax} = \mathbf{0}_m$ must be in the column space of $\mathbf{X}$. Let the submatrix of the top r rows of $\mathbf{x}$ be $\mathbf{x}_1$ and the submatrix of the bottom $n-r$ rows of $\mathbf{x}$ be $\mathbf{x}_2$.

$$\mathbf{Ax} = \begin{pmatrix} \mathbf{A}_1 & \mathbf{A}_1\mathbf{B} \end{pmatrix} \begin{pmatrix} \mathbf{x}_1 \\ \mathbf{x}_2 \end{pmatrix} = \mathbf{A}_1\mathbf{x}_1 + \mathbf{A}_1\mathbf{B}\mathbf{x}_2 = \mathbf{A}_1(\mathbf{x}_1 + \mathbf{B}\mathbf{x}_2) = \mathbf{0}_m$$

Since the columns of $\mathbf{A}_1$ are linearly independent, $\mathbf{A}_1\mathbf{u} = \mathbf{0}_r$ implies $\mathbf{u} = \mathbf{0}_r$, thus

$$\mathbf{u}_1 + \mathbf{B}\mathbf{u}_2 = \mathbf{0}_r$$

$$\mathbf{u} = \begin{pmatrix} -\mathbf{B} \\ \mathbf{I}_{n-r} \end{pmatrix} \mathbf{u}_2 = \mathbf{X}\mathbf{u}_2$$

This proves that any vector $\mathbf{u}$ that is a solution of $\mathbf{Ax} = \mathbf{0}$ must be a linear combination of the $n-r$ special solutions given by the columns of $\mathbf{X}$. And we have already seen that the columns of $\mathbf{X}$ are linearly independent.

Hence, the column space of $\mathbf{X}$ is the null space of $\mathbf{A}$. Therefore, the nullity of $\mathbf{A}$ is $n-r$. Since r is the rank of $\mathbf{A}$, it follows that $\text{rank}(\mathbf{A}) + \ker(\mathbf{A}) = n$. □

xi Rank-left nullity theorem

For any $m \times n$ matrix $\mathbf{A}$:

$$\text{rank}(\mathbf{A}) + \dim\left(\ker\left(\mathbf{A}^\mathsf{T}\right)\right) = m$$

Proof. Apply rank-nullity theorem on A^T. □

xii Column space test for consistency of linear system

A system of linear equations is consistent iff its constant term vector is in the column space of its coefficient matrix.

Proof. By definition, obviously. □

Thus, a homogeneous linear system is always consistent.

xiii Rank and number of pivot columns

For a matrix A and its reduced row echelon form R, the followings are equal:

- The rank of A (or R)
- The number of pivot positions of A (or R)

- The number of pivot columns of A (or R)
- The number of pivot rows of A (or R)
- The number of nonzero rows of R
- The number of basic variables in the general solution of any consistent system of linear equations whose coefficient matrix is A (or R)

xiv Nullity and number of non-pivot columns

For a matrix A and its reduced row echelon form R, the followings are equal:

- The nullity of A (or R)
- The number of non-pivot columns of A (or R)
- The number of free variables in the general solution of any consistent system of linear equations whose coefficient matrix is A (or R)

xv Left nullity and number of non-pivot rows

For a matrix A and its reduced row echelon form R, the followings are equal:

- The left nullity of A (or R)
- The number of non-pivot rows of A (or R)
- The number of zero rows of R
- The number of free variables in the general solution of any consistent system of linear equations whose coefficient matrix is $A^\top$ (or $R^\top$)

xvi Full rank (滿秩) and rank deficiency (秩虧)

For an $m \times n$ matrix **A**, if

$$\text{rank}(\mathbf{A}) = \min(m, n)$$

A is called to have full rank; otherwise it is called to be rank-deficient (欠秩或秩虧的).

The rank deficiency of a $m \times n$ matrix **A** is defined as

$$\min(m, n) - \text{rank}(\mathbf{A})$$

xvii Property of rank

- For any $m \times n$ matrix **A**:

$$\text{rank}(\mathbf{A}) \leq \min(m, n)$$

- A square matrix is invertible if and only if it has full rank.
- A matrix is a zero matrix if and only if its rank is zero.

3 Matrix Analysis (矩陣分析)

I Eigenvector or characteristic vector (特徵向量) and characteristic polynomial (特徵多項式)

i Eigenvector or characteristic vector (特徵向量) and eigenvalue (本徵值、固有值、特徵值或特徵根)

Given a linear transformation $L : V \to V$, an eigenvector of it is defined as any nonzero vector $v \in V$ such that:

$$\mathbf{A}v = \lambda v,$$

where λ is a scalar, which is called the eigenvalue of $\mathbf{A}$ or the eigenvalue of v.

Obviously, v is an eigenvector implies av is an eigenvector of the same eigenvalue for any nonzero scalar a.

ii Characteristic polynomial

For an $n \times n$ matrix $\mathbf{A}$, the characteristic polynomial of it is defined as

$$p_{\mathbf{A}}(\lambda) := \det\bigl(\lambda \mathbf{I}_n - \mathbf{A}\bigr),$$

where λ is a scalar variable.

A scalar λ is an eigenvalue of $\mathbf{A}$ if and only if

$$p_{\mathbf{A}}(\lambda) = 0.$$

iii Property

- The number of distinct eigenvalues of a square matrix is less than or equal to the number of linearly independent eigenvectors.
- The number of distinct eigenvalues of a $n \times n$ matrix is less than or equal to n.
- The algebraic multiplicity of a root λ of the characteristic polynomial of a square matrix is greater than or equal to the number of linearly independent eigenvectors of it whose eigenvalue is λ.

iv Cayley–Hamilton theorem (凱萊–哈密頓定理)

For a square matrix $\mathbf{A}$ and its characteristic polynomial $p_{\mathbf{A}}$, the Cayley–Hamilton theorem states that:

$$p_{\mathbf{A}}(\mathbf{A}) = 0$$

v Diagonalization (對角化)

The diagonalization of an $n \times n$ matrix $\mathbf{A}$ is the construction of an $n \times n$ matrix $\mathbf{P}$ and an $n \times n$ diagonal matrix $\mathbf{D}$ such that:

$$\mathbf{A} = \mathbf{P}^{-1}\mathbf{D}\mathbf{P}$$

An $n \times n$ matrix can be diagonalized if it has n distinct eigenvalues.

The steps of the diagonalization of an $n \times n$ matrix **A** is as follows:

1. For each eigenvalue λ of the matrix **A**, solve an eigenvector v:

$$(\mathbf{A} - \lambda \mathbf{I})\mathbf{v} = 0.$$

2. Compose matrix **P** with each eigenvector of **A** as a column (or row).

Common application:

$$\mathbf{A}^n = (\mathbf{P}^{-1}\mathbf{D}\mathbf{P})^n = \mathbf{P}^{-1}\mathbf{D}^n\mathbf{P}$$

vi Definiteness (正定性)

Let $A \in \mathbb{R}^{n \times n}$. A is called

- Positive definite: iff

$$\forall \mathbf{x} \in \mathbb{R}^n, \mathbf{x} \neq 0 : \ \mathbf{x}^\top A \mathbf{x} > 0.$$

- Negative definite: iff

$$\forall \mathbf{x} \in \mathbb{R}^n, \mathbf{x} \neq 0 : \ \mathbf{x}^\top A \mathbf{x} < 0.$$

- Positive semi-definite: iff

$$\forall \mathbf{x} \in \mathbb{R}^n, \mathbf{x} \neq 0 : \ \mathbf{x}^\top A \mathbf{x} \geq 0.$$

- Negative semi-definite: iff

$$\forall \mathbf{x} \in \mathbb{R}^n, \mathbf{x} \neq 0 : \ \mathbf{x}^\top A \mathbf{x} \leq 0.$$

- Indefinite: otherwise, that is,

$$\exists \mathbf{x} \in \mathbb{R}^n, \mathbf{x} \neq 0 : \ \mathbf{x}^\top A \mathbf{x} > 0$$

  and

$$\exists \mathbf{x} \in \mathbb{R}^n, \mathbf{x} \neq 0 : \ \mathbf{x}^\top A \mathbf{x} < 0.$$

vii Eigenvalues Test of Definiteness for Symmetric Hessians of Scalar Fields

For a symmetric matrix A,

- A is positive definte if all eigenvalues of it are positive.
- A is negative definte if all eigenvalues of it are negative.
- A is positive semidefinte if all eigenvalues of it are nonnegative and at least one eigenvalue of it is positive.
- A is negative semidefinte if all eigenvalues of it are nonpositive and at least one eigenvalue of it is negative.
- A is neutral semidefinte if all eigenvalues of it are zero.
- A is indefinte otherwise.

II Generalized Definition of Inverse

i One-sided inverse (單邊逆)

- Right inverse (右逆): For any $n \times m$ matrix A with rank $\operatorname{rank}(A) = n$, there exists an $m \times n$ matrix A_{R}^{-1}, called the right inverse of **A**, such that

$$AA_{\mathrm{R}}^{-1} = I_n,$$

  where I_n is the $n \times n$ identity matrix.

- Left inverse (左逆): For any $n \times m$ matrix A with rank $\operatorname{rank}(A) = m$, there exists an $m \times n$ matrix A_{L}^{-1}, called the left inverse of **A**, such that

$$A_{\mathrm{L}}^{-1}A = I_m,$$

  where I_m is the $m \times m$ identity matrix.

Properties:

- A matrix doesn't necessarily have a right inverse and doesn't necessarily have a left inverse.
- A square matrix has a right inverse if and only if it has left inverse if and only if it is invertible.
- For any invertible matrix, there exists only one right inverse and one left inverse, which are its multiplicative inverse.

ii Generalized inverse or g-inverse (廣義逆)

A matrix $A^{\mathrm{g}} \in \mathbb{C}^{n \times m}$ is a generalized inverse of a matrix $A \in \mathbb{C}^{m \times n}$ if:

$$AA^{\mathrm{g}}A = A.$$

Properties:

- For all matrices, there exists at least one generalized inverse, but it's not necessarily unique.
- For any invertible matrix, there exists only one generalized inverse, which is its multiplicative inverse.

iii Reflexive generalized inverse (自反廣義逆)

A reflexive generalized inverse of a matrix **A** is defined as a matrix $\mathbf{A}^g$ such that:

$$\mathbf{A}\mathbf{A}^g\mathbf{A} = \mathbf{A}$$

$$\mathbf{A}^g\mathbf{A}\mathbf{A}^g = \mathbf{A}^g$$

Properties:

- For all matrices, there exists at least one reflexive generalized inverse, but it's not necessarily unique.
- For any invertible matrix, there exists only one reflexive generalized inverse, which is its multiplicative inverse.

iv Moore–Penrose inverse (摩爾—彭若斯廣義逆) or pseudoinverse (偽逆或擬反)

A Moore–Penrose inverse of a matrix $\mathbf{A}$ is defined as a matrix $\mathbf{A}^+$ such that:

$$\mathbf{A}\mathbf{A}^+\mathbf{A} = \mathbf{A}$$

$$\mathbf{A}^+\mathbf{A}\mathbf{A}^+ = \mathbf{A}^+$$

$$\left(\mathbf{A}\mathbf{A}^+\right)^* = \mathbf{A}\mathbf{A}^+$$

$$\left(\mathbf{A}^+\mathbf{A}\right)^* = \mathbf{A}^+\mathbf{A}$$

Properties:

- For all matrices, there exists only one Moore–Penrose inverse.
- For any invertible matrix, there exists only one Moore–Penrose inverse, which is its multiplicative inverse.

4 Transformation Matrices

I Groups

i General linear group

The general linear group of degree n over some field F is the set of $n \times n$ invertible matrices over F, together with the operation of matrix multiplication, denoted as $\mathrm{GL}(n, F)$ or $\mathrm{GL}_n(F)$, or simply $\mathrm{GL}(n)$ if the field is understood. It is a topological group and a Lie group under operation $+$.

The Lie algebra of a general linear group is denoted as the symbol of the general linear group with GL replaced with $\mathfrak{gl}$.

ii Orthogonal group or general orthogonal group

The orthogonal group or general orthogonal group of degree n over some field F is the set of $n \times n$ invertible matrices over F such that the inverse equals the transpose, together with the operation of matrix multiplication, denoted as $\mathrm{O}(n, F)$ or $\mathrm{O}_n(F)$, or simply $\mathrm{O}(n)$ if the field is understood. An element of an orthogonal group is called an orthogonal matrix. It is a topological group and a Lie group under operation $+$.

iii Special orthogonal group or rotation group

The special orthogonal group or rotation group of degree n over some field F is a subgroup of the orthogonal group of degree n over some field F such that for any matrix in it, the determinant is 1, that is, it preserves the orientation, denoted as $\mathrm{SO}(n, F)$ or $\mathrm{SO}_n(F)$, or simply $\mathrm{SO}(n)$ if the field is understood. An element of a special orthogonal group is called an rotational matrix. It is a topological group and a Lie group under operation $+$.

The Lie algebra of a special orthogonal group is denoted as the symbol of the special orthogonal group with SO replaced with $\mathfrak{so}$.

II Transformation matrices in two-dimensional space

i Degenerate matrices

Degenerate to 1 dimension ($(a \vee b) \wedge (d \vee e) \neq 0$):

$$\begin{bmatrix} a & b \\ ca & cb \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} ax + by \\ c(ax + by) \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 \\ d & e \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ dx + ey \end{bmatrix}$$

Degenerate to 0 dimension:

$$\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

ii Stretching (伸縮/縮放/拉伸)

A stretch is a linear transformation which enlarges all distances in a particular direction by a constant factor k but does not affect distances in the perpendicular direction. (Note that if $k > 1$, then this really is a ”stretch”; if $k < 1$, it is technically a ”compression”, but we still call it a stretch. Also, if $k = 1$, then the transformation is an identity, i.e. it has no effect)

The matrix associated with a stretch by a factor h along the x-axis and a factor k along the y-axis is given by:

$$\begin{bmatrix} h & 0 \\ 0 & k \end{bmatrix}$$

iii Shearing (推移)

A shearing in the x-direction dependent on the y coordination by a factor k is given by:

$$\begin{bmatrix} 1 & k \\ 0 & 1 \end{bmatrix}$$

A shearing in the y-direction dependent on the x coordination by a factor k is given by:

$$\begin{bmatrix} 1 & 0 \\ k & 1 \end{bmatrix}$$

iv Rotation (旋轉)

A rotation by an angle θ counterclockwise about the origin is given by:

$$\begin{bmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{bmatrix}$$

v Reflection (鏡射/反射)

A reflection about a line that passes the original and with $(a, b) = \sqrt{a^2 + b^2}(\cos\theta, \sin\theta)$ being a vector in the direction of it is given by:

$$\begin{bmatrix} \cos(2\theta) & \sin(2\theta) \\ \sin(2\theta) & -\cos(2\theta) \end{bmatrix} = \frac{1}{a^2 + b^2} \begin{bmatrix} a^2 - b^2 & 2ab \\ 2ab & b^2 - a^2 \end{bmatrix}$$

vi Orthogona projection (正射影)

An orthogonal projection about a line that passes the original and with $(a, b) = \sqrt{a^2 + b^2}(\cos\theta, \sin\theta)$ being a vector in the direction of it is given by:

$$\begin{bmatrix} \cos^2(\theta) & \cos(\theta)\sin(\theta) \\ \cos(\theta)\sin(\theta) & \sin^2(\theta) \end{bmatrix} = \frac{1}{a^2 + b^2} \begin{bmatrix} a^2 & ab \\ ab & b^2 \end{bmatrix}$$

vii Exmaple 1

Question: The cricle C : $x^2 + y^2 = 1$ is transformed by $\mathbf{A} = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}$ to obtain a new curve C′. Find the equation of C′ and its area.

Answer:

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \mathbf{A} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 2x \\ 3y \end{bmatrix}$$

$$\begin{cases} x' = 2x \\ y' = 3y \end{cases} \Rightarrow \begin{cases} x = \dfrac{x'}{2} \\ y = \dfrac{y'}{3} \end{cases}$$

$$\left(\frac{x'}{2}\right)^2 + \left(\frac{y'}{3}\right)^2 = 1 \Rightarrow \frac{x'^2}{4} + \frac{y'^2}{9} = 1$$

$$\text{C}' : \frac{x'^2}{4} + \frac{y'^2}{9} = 1$$

The area of C′ is $\pi \cdot 2 \cdot 3 = 6\pi$. Since $\det(\mathbf{A}) = 6$, and the area of C is π, the area of C′ equals $\det(\mathbf{A})$ multiplied by the area of C.

viii Exmaple 2

Question: The line **L** : $4x - 3y = 5$ is transformed in the x-direction dependent on the y coordination by twice and then in the y-direction dependent on the x coordination by negative theee times to obtain a new line **L**′. Find the equation of **L**′.

Answer:

$$\mathbf{A} = \begin{bmatrix} 1 & 2 \\ -3 & 1 \end{bmatrix}$$

$$\mathbf{A}^{-1} = \frac{1}{7}\begin{bmatrix} 1 & -2 \\ 3 & 1 \end{bmatrix}$$

$$\begin{bmatrix} x \\ y \end{bmatrix} = \mathbf{A}^{-1}\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} \frac{x' - 2y'}{7} \\ \frac{3x' + y'}{7} \end{bmatrix}$$

$$\mathbf{L}' : 4\frac{x - 2y}{7} + 3\frac{3x + y}{7} = 5$$

$$\mathbf{L}' : 5x + 11y + 35 = 0$$

ix Example 3

Question: Reflect the point P$(5, 2)$ over the line $\mathbf{M} : x - y + 1 = 0$ to obtain a new point Q. Find Q.

Answer:

Shift one unit to the right: **M** becomes $\mathbf{M}' : x - y = 0$ and C becomes C$'(6, 2)$.

The reflection matrix is:

$$\mathbf{A} = \begin{bmatrix} 0 & 1 \, 1 & 0 \end{bmatrix}$$

$$\mathrm{D}' = \mathbf{A}\mathrm{C}' = (2, 6)$$

Shift one unit to the left: D$(1, 6)$

x Example 4

Question: The line $x + y - 2 = 0$ is transformed by $\mathbf{A} = \begin{bmatrix} a & b \\ 1 & 2 \end{bmatrix} \in \mathbb{R}^{2\times 2}$ to obtain a new line $2x + 3y - 4 = 0$. Find a and b.

Answer:

$$\mathbf{A}\begin{bmatrix} x \\ -x + 2 \end{bmatrix} = \begin{bmatrix} x' \\ \frac{-2x' + 4}{3} \end{bmatrix}$$

$$x - 2x + 4 = \frac{-2x' + 4}{3}$$

$$x' = \frac{3}{2}x - 4$$

$$ax - bx - 2b = x' = \frac{3}{2}x - 4$$

$$(a, b) = (-\frac{1}{2}, -2)$$

III Transformation matrices in three-dimensional space

i Stretching (伸縮/縮放/拉伸)

The matrix associated with a stretch by a factor h along the x-axis, a factor k along the y-axis, and a factor w along the z-axis is given by:

$$\begin{bmatrix} h & 0 & 0 \\ 0 & k & 0 \\ 0 & 0 & w \end{bmatrix}$$

ii Shearing (推移)

A shearing in the x-direction dependent on the y coordination by a factor k is given by:

$$\begin{bmatrix} 1 & k & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

A shearing in the x-direction dependent on the z coordination by a factor k is given by:

$$\begin{bmatrix} 1 & 0 & k \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

A shearing in the y-direction dependent on the x coordination by a factor k is given by:

$$\begin{bmatrix} 1 & 0 & 0 \\ k & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

A shearing in the y-direction dependent on the z coordination by a factor k is given by:

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & k \\ 0 & 0 & 1 \end{bmatrix}$$

A shearing in the z-direction dependent on the x coordination by a factor k is given by:

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ k & 0 & 1 \end{bmatrix}$$

A shearing in the z-direction dependent on the y coordination by a factor k is given by:

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & k & 1 \end{bmatrix}$$

iii Rotation (旋轉)

A rotation by an angle θ counterclockwise about a line that passes the original and with (a, b, c) being a vector in the direction of it is given by:

$$\frac{1}{a^2+b^2+c^2}\begin{bmatrix} a^2(1-\cos\theta)+\cos\theta & ab(1-\cos\theta)-c\sin\theta & ac(1-\cos\theta)+b\sin\theta \\ bc(1-\cos\theta)+c\sin\theta & b^2(1-\cos\theta)+\cos\theta & bc(1-\cos\theta)-a\sin\theta \\ ac(1-\cos\theta)-b\sin\theta & bc(1-\cos\theta)+a\sin\theta & c^2(1-\cos\theta)+\cos\theta \end{bmatrix}$$

iv Rodrigues' rotation formula

If $\mathbf{v}, \mathbf{k} \in \mathbb{R}^3$ and $\mathbf{k}$ is a unit vector describing an axis of rotation about which $\mathbf{v}$ rotates by an angle θ according to the right hand rule, the Rodrigues formula for the rotated vector $\mathbf{v}_r$ is

$$\mathbf{v}_r = \mathbf{v}\cos\theta + (\mathbf{k}\times\mathbf{v})\sin\theta + \mathbf{k}(\mathbf{k}\cdot\mathbf{v})(1-\cos\theta).$$

The skew-symmetric matrix $\mathbf{K}$ in Lie algebra $\mathfrak{so}(3)$ that generates the one-parameter subgroup of $\mathrm{SO}(3)$:

$$\mathbf{R}(\theta) = \exp(\theta\mathbf{K}),$$

such that

$$\mathbf{v}_r = \mathbf{R}(\theta)\mathbf{v},$$

is given by

$$\mathbf{K}\mathbf{v} = \mathbf{k}\times\mathbf{v}, \quad \forall\mathbf{v}\in\mathbb{R}^3,$$

and

$$\mathbf{R}(\theta) = \mathbf{I} + \sin\theta\mathbf{K} + (1-\cos\theta)\mathbf{K}^2,$$

where $\mathbf{I}$ is the 3×3 identity matrix.

v Reflection (鏡射/反射)

A reflection about a plane $ax+by+cz=0$ is given by:

$$\mathbf{I} - \frac{1}{a^2+b^2+c^2}\begin{bmatrix} 2a^2 & 2ab & 2ac \\ 2ab & 2b^2 & 2bc \\ 2ac & 2bc & 2c^2 \end{bmatrix},$$

where $\mathbf{I}$ is the 3×3 identity matrix.

vi Orthogona projection (正射影)

An orthogonal projection about a plane $ax+by+cz=0$ is given by:

$$\mathbf{I} - \frac{1}{a^2+b^2+c^2}\begin{bmatrix} a^2 & ab & ac \\ ab & b^2 & bc \\ ac & bc & c^2 \end{bmatrix}$$

where $\mathbf{I}$ is the 3×3 identity matrix.